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A thin robot made of flexible electronics for in-situ machining and inspection of large structures

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In-situ machining and inspection of large structures drive demand, but conventional robots are typically heavy and bulky, potentially damaging fragile materials and limiting effectiveness in confined spaces or on complex surfaces. Here, we present a thin (970 μm), lightweight (19.7 g), and flexible robot powered by a two-degree-of-freedom electrostatic actuator made of flexible circuits. The actuator enables motion with 21.7 μm repeatability, a maximum speed of 290 mm/s, and a superposable output force up to 1.07 N. The robot follows arbitrary trajectories and operates reliably on diverse surfaces, including flat, curved, and vertical substrates, through electrostatic adhesive feet. It carries payloads up to 4.06 times its own weight and tolerates impact from a 3 kg weight. We demonstrate multifunctional in-situ operation through narrow-gap inspection, curved-surface grinding, and microscale electrical discharge machining. This establishes a thin, flexible platform for in-situ tasks in constrained and sensitive environments.

Large industrial workpieces, such as aircraft and ships, require frequent machining and inspection for cleaning, maintenance, repair, or assembly[1,2]. However, their large dimensions and complex geometries are often incompatible with traditional machine tools, which typically accommodate workpieces no larger than 1 meter in size[3]. One solution is the use of oversized machine tools, but this approach faces significant challenges related to system size, fabrication complexity, precision limitations, and high cost[4,5]. An increasingly promising alternative is the deployment of mobile machine robots capable of performing in-situ inspection and machining tasks[6].

Over the past few decades, numerous studies have explored the development of machine robots for in-situ applications[7,8]. A typical early prototype comprises a wheeled platform integrated with either a robotic arm or a parallel kinematic mechanism to perform machining or inspection tasks[9–12]. As an alternative, legged robotic platforms have been proposed to enhance mobility and adaptability on complex surfaces[13,14], such as the miniature machine tool named Free-Leg Hexapod[15–17], and a six-parallel-legged walking robot[18]. However, existing robotic platforms predominantly utilize rigid metal components, resulting in substantial weight (exceeding 7 kg) and large form factors (height over 0.35 m)[15,19–23]. Such characteristics pose significant risks to delicate surfaces during operation. For example, concentrated loads can cause microcracks in composite aircraft skins, potentially leading to delamination or fiber breakage[24,25].

Similarly, the weight of heavy robots may deform thin-shell structures or damage protective coatings, increasing the risk of corrosion[26,27]. Additionally, their bulky size limits access to confined spaces, reducing their effectiveness in many in-situ machining and inspection tasks[28,29]. Moreover, most machine robots can only move on horizontal surfaces, but cannot freely move on vertical or curved surfaces, which are very common in the large workpieces of aircraft and ships[30,31]. Conventional inspection robots utilize magnetic adhesion[32] or vacuum suction cups[33,34] to traverse vertical planes[35,36] and curved[37,38] surfaces of large structures; they still rely on rigid components, resulting in considerable weight (exceeding 0.45 kg) and height (greater than 0.05 m)[32,33]. The limited motion accuracy of these

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inspection robots[35] impedes the integration of in-situ machining capabilities. In recent years, lightweight flexible actuators[39,40] and adhesive devices[41,42] have been proposed to enable microrobot traversal of complex surfaces. However, these solutions suffer from large trajectory tracking errors (Supplementary Table 1), short machining strokes (Supplementary Table 2), and some thickness limitations (Supplementary Table 3 and Supplementary Fig. 1). In-situ machining requires both robot-level trajectory tracking and actuator-level high-precision, long-stroke 2D tool motion, which remains challenging for current microrobots. As a result, there remains a need for a robotic platform that combines positioning capability, compact form factor, and adaptability to enable expanded in-situ machining and inspection in confined and geometrically complex environments.

To overcome the aforementioned challenges, this study proposes a two-degree-of-freedom (2-DOF) high-precision thin robot (Fig. 1a) by integrating electrostatic film actuators for both in-situ inspection and machining tasks. These electrostatic actuators, fabricated using flexible circuit technology, operate based on electrostatic forces generated between designed microscale electrodes embedded within polymer insulation layers[43,44]. Robots built from flexible electronics offer advantages such as lower cost, enhanced environmental sustainability, and efficient fabrication processes[45,46]. Whereas conventional electrostatic actuators typically enable motion along a single axis, we developed a 2-DOF electrostatic film actuator with a two-layer, orthogonally distributed micrometer-scale electrode architecture. This design enables decoupled planar motion along both the X and Y directions and supports superposable output force through multilayer stacking. The resulting robot exhibits a profile of 970 μm, over an order of magnitude thinner than existing platforms (Fig. 1b and Supplementary Table 4), allowing it to access narrow gaps that were previously inaccessible. It is also lightweight (19.7 g), more than three orders of magnitude

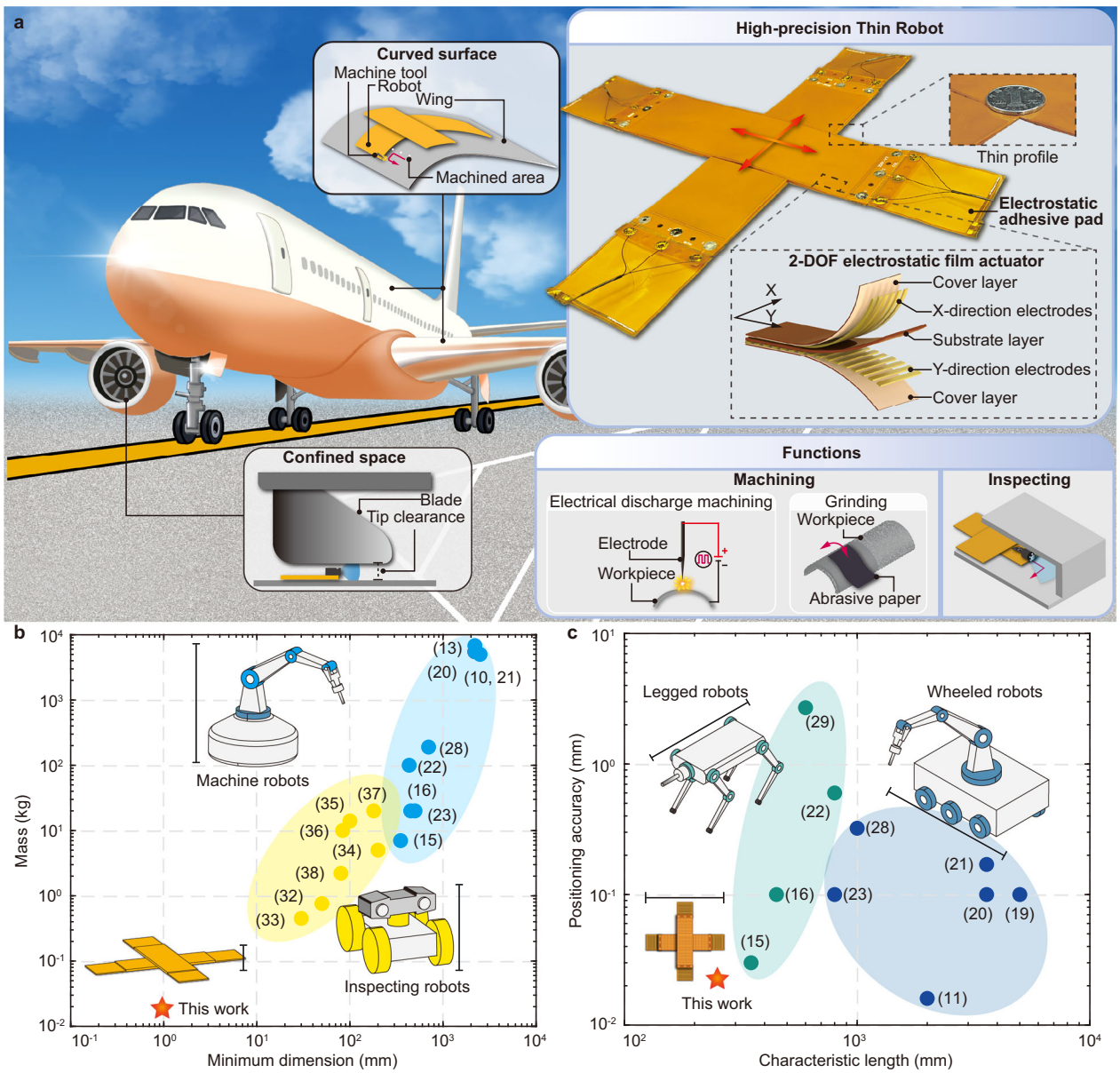


Fig. 1 | Design concept of the high-precision thin robot. a Schematic of the design and application of the robot. The robot features a flexible and thin design, enabled by a two-degree-of-freedom (2-DOF) electrostatic film actuator and four electrostatic adhesive feet. It performs in-situ machining and inspection tasks on curved surfaces and in confined spaces using changeable tools. Comparison of the minimum dimension versus mass **b** and characteristic length versus positioning accuracy **c** of the robot with other inspecting and machine robots, including wheeled and legged platforms.

lighter than most conventional robots, enabling safe operation on fragile and sensitive surfaces. Despite its compliant structure, the robot demonstrates a maximum output force of 1.07 N, a speed of up to 290 mm/s, and a linear positioning accuracy of 23.1 μm. With comparable positioning capability to prior machining robots (Fig. 1c), the robot can follow arbitrary trajectories, including rectangular, circular, and alphanumeric paths. It can operate across both flat and vertical substrates composed of diverse materials, such as metal, paper, wood, rubber, and acrylic. It can also carry payloads up to 4.06 times its own weight. Benefiting from its flexible structure, the robot remains operational after sustaining impact deformation from a 3 kg load (over 152 times its own weight) and can conform to curved surfaces with a 560 mm radius of curvature. In this work, we demonstrate the robot's functionality through representative applications, including inspection within a 7 mm-high narrow gap with 3D reconstruction of internal ribbed structures, flexible grinding on curved workpieces, and electrical discharge machining (EDM) of a 3×3 array on an aluminum plate.

Results

Design and modeling of the 2-DOF electrostatic film actuator

The traditional electrostatic film actuator with single-layer parallel electrodes performs only 1 DOF movement[47,48] (Supplementary Fig. 2a). To provide electrostatic film actuators with an additional DOF, we designed the electrostatic film actuator with a two-layer electrode layout. The electrodes within the same layer are parallel, but the electrodes in the two layers are orthogonal. Specifically, the top electrode layer contains three-phase parallel electrodes along the X-direction (with phases 1–3), while the bottom electrode layer features another set of electrodes along the Y-direction (with phases 4–6), as shown in Fig. 2a. The bus line of each electrode phase is connected to the other layer through vias. This novel two-layer electrode layout gives each actuation film of the electrostatic film actuator two actuation interfaces with perpendicular directions of motion, referred to as the 2-DOF electrostatic film actuator.

The minimum actuation unit of the 2-DOF electrostatic film actuator consists of two stators and one slider, as shown in Fig. 2a. The contact surfaces between the slider and stators 1 and 2 form actuation interfaces 1 and 2, where two pairs of three-phase alternating voltages drive the slider to move in both directions along the X and Y axes, thereby enabling 2-DOF planar motion. Additionally, benefiting from the dual actuation interfaces, this electrostatic film actuator can increase its output force by stacking additional films. This stacking scales the bending strain and can reduce conformability on curved substrates, reflecting a tradeoff between output force and conformability. According to the linear driving force models[49,50], the net output force of the 2-DOF electrostatic film actuator in the XY plane can be calculated as:

$$\mathbf{F} = K_x v_a^2 \sin(\theta_{ex} - 2\phi_x)\mathbf{i} + K_y v_a^2 \sin(\theta_{ey} - 2\phi_y)\mathbf{j}, \tag{1}$$

where $K_x = (m/2 - 1)k_f A_e$ and $K_y = (m/2)k_f A_e$ are the force constants with units of $N{\cdot}V^{-2}$ for the output forces in the X and Y directions of the stacked 2-DOF actuator, and m represents the even number of actuation films used in stacking, with both the slider and stator groups consisting of $m/2$ films. The number of actuation interfaces in the X direction is one fewer than in the Y direction. k_f is a force constant with units of $N{\cdot}m^{-2}{\cdot}V^{-2}$. A_e is the effective actuation area of the electrodes and is identical for the X and Y directions. v_a is the amplitude of the actuation voltages. $\theta_{ex} = 2\pi x_{sl}/3p$ and $\theta_{ey} = 2\pi y_{sl}/3p$ represent the electric angles of the slider along the X and Y directions, respectively, where x_{sl} and y_{sl} denote the planar positions of the slider, and p is the sum of the width and the space of electrodes called pitch. ϕ_x and ϕ_y are the excitation phases of voltages applied to the X and Y-direction electrodes.

Similarly, the net speed of the slider is:

$$\mathbf{v} = 6pf_x\mathbf{i} + 6pf_y\mathbf{j}, \tag{2}$$

where f_x and f_y are the frequencies of voltages applied to the X and Y directions electrodes. Thus, this actuator's velocity can be accurately controlled by open-loop control.

The net displacement of the slider is:

$$\mathbf{d} = \frac{3p}{2\pi}\left(2\phi_x - \sin^{-1}\frac{F_x}{K_x v_0^2}\right)\mathbf{i} + \frac{3p}{2\pi}\left(2\phi_y - \sin^{-1}\frac{F_y}{K_y v_0^2}\right)\mathbf{j}, \tag{3}$$

where F_x and F_y are the output forces of the actuator in the X and Y directions. According to the equation, when the loads $F_{lx} \le K_x v_0^2$ and $F_{ly} \le K_y v_0^2$, the equilibrium positions can be shifted by changing the excitation phases ϕ_x and ϕ_y, enabling open-loop control of the actuator.

Moreover, the actuator can operate on curved surfaces after bending. If the bending axis is in the X direction, the driving force in the X direction remains the same, while the driving force in the Y direction varies with the bending radius since the electrodes in the slider and stator are not aligned at the same electrical angle positions anymore. Based on the previous study[51], the driving force per unit area in the Y direction F_{yb} can be calculated by the driving force per unit area of microelement f_{yb}:

$$\frac{dt}{d\varphi} = rf_{yb} - \mu_a t, \tag{4}$$

where t is the tensile force, φ is the bending angle, r is the radius of curvature corresponding to the curved section, and μ_a is the friction coefficient between actuation films. The schematic of the bending structure and the result of output forces are detailed in Supplementary Note 1 and Supplementary Fig. 2b, c.

Here, we employed the electrostatic adhesive pad as feet. The model for tangential adhesive force is:

$$F_t = \mu_{ad}F_n = \mu_{ad}k_{ad}v_{ad}^2, \tag{5}$$

where F_n is the normal adhesive force, μ_{ad} is the friction coefficient between the adhesive pad and substrate, k_{ad} is the force constant with units of $N{\cdot}V^{-2}$ for the adhesive force based on the materials and dimensions of the electrostatic adhesive pad, v_{ad} is the amplitude of the adhesive voltage.

As shown in Supplementary Fig. 3a, the simulations were carried out for three types of layouts to examine the output force of the slider on one actuation interface: Layout 1, where only the electrode layers 2 and 3 near the actuation interface were present and enabled; Layout 2, where electrode layers 1 and 4 were grounded; Layout 3, where all electrode layers had voltage applied. In the simulation, the electrostatic film actuator was placed inside an air-filled box, and the sides of the region were set to zero potential to simulate infinitely distant boundaries. The materials and primary parameters of each film are listed in Supplementary Table 5, based on the manufacturing process used in this work, with both the electrode width and spacing set to 100 μm.

To explore the interference between different electrode layers in the 2-DOF electrostatic film actuator with the two-layer orthogonally distributed electrodes, the output forces of the actuators with different layouts (Supplementary Fig. 3a) were calculated using the finite element analysis model. By applying driving voltages to the electrodes of each phase (layout 3) in the simulation, as shown in Supplementary Fig. 3b, the electric potential distribution of the 2-DOF electrostatic film actuator was obtained.

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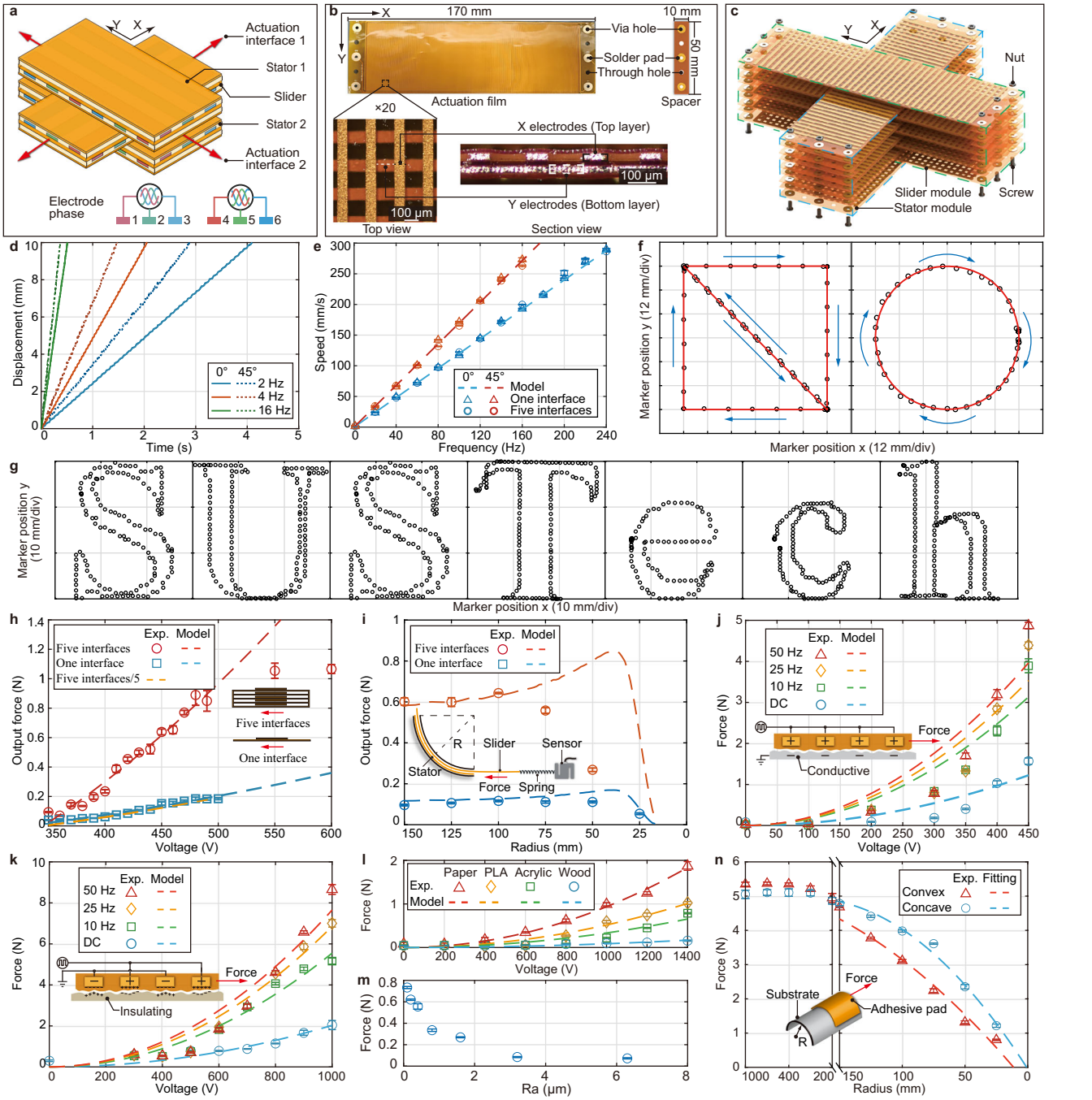


Fig. 2 | Design and performance of the 2-DOF electrostatic film actuator and the electrostatic adhesive pad. a The design and principle of the 2-DOF electrostatic film actuator with two-layer orthogonally distributed electrodes. **b** Images of the actuation film and its electrode details in different copper layers. **c** Schematic of the 2-DOF electrostatic film actuator assembly. **d** Displacements in different directions and driving voltage frequencies. **e** Speed in different directions, driving voltage frequencies, and the number of actuation interfaces. **f** Trajectory tracking performance of rectangular and diagonal paths, and a circular path. **g** Demonstration of complex lettering trajectory spelling SUSTech, corresponding to the authors' institutional affiliation. **h** Measured output force of actuators with one and five actuation interfaces, showing the force superposition behavior under multilayer actuation. **i** Output force of actuators with one and five actuation interfaces under varying bending curvatures, represented by different curvature radii. Lateral adhesive forces of the electrostatic adhesive pad on **j** conductive and **k** rubber substrates as functions of driving voltage and frequency. Lateral adhesive forces were measured at a driving frequency of 50 Hz. **l** Measured on various insulating substrates (paper, PLA, acrylic, and wood). **m** Measured on substrates with different surface roughness. **n** Measured on convex and concave conductive surfaces. The data are presented as mean values ± standard deviation (SD) for the number of trials $n = 7$.

This further yielded the relationship between the output force and the excitation phase at different electric angles within one pitch, as shown in Supplementary Fig. 3c. The relationship between the peak output forces and the electric angle for different layouts is shown in Supplementary Fig. 3d, where F_{pmin} represents the maximum output force achievable during motion. The results show no significant variation in output force across the three electrode layouts, indicating minimal interference between layers and confirming that actuation in the X and Y directions is effectively decoupled. This decoupling facilitates more straightforward, precise control of both velocity and displacement.

Implementation of the robot

The robot was composed of a 2-DOF electrostatic film actuator and four electrostatic adhesive pads. The manufacturing process of the actuation films for the actuator is illustrated in Supplementary Fig. 4 and detailed in Methods. As shown in Fig. 2b, inside the actuation film, the copper electrodes were as thin as 78 μm in width (122 μm spacing). The longitudinal electrodes and transverse electrodes were distributed at the top copper layer and the bottom copper layer, respectively. Each actuation film measured 170 mm in length and 50 mm in width, with a thickness of 96 μm and a weight of 1.7 g. The slider/stator module was composed of multiple films that were connected together on the two edges by solder or screws (Fig. 2c).

In this work, both the stator module and the slider module contained five actuation films. The films were separated by spacers (50 mm × 10 mm × 96 μm) to ensure consistent spacing between the films, as shown in Fig. 2c. Glass beads (20–30 μm in diameter) were evenly applied to the surfaces of each actuation film as a lubricant. The assembled prototype had a footprint area of approximately 170 × 170 mm^2, and the number of actuation interfaces for the X-direction and Y-direction was 4 and 5, respectively. The weight of the actuation films and spacers was approximately 16.8 g, and the solder (or the screws and nuts) weighed around 0.8 g (or 2.4 g), making the total weight of the actuator (excluding wires) approximately 17.6 g (or 19.2 g).

Electrostatic adhesive pads (50 × 50 mm^2) were utilized as adhesive feet for the robot, enabling independent control of friction forces between the slider/stator and the substrate (Supplementary Fig. 5a). Each electrostatic adhesive pad possessed interdigitated electrodes (100 μm width, 100 μm spacing) with an insulation cover. The electrostatic adhesive foot was connected to the end of the actuation module via a flexible joint, which was made by the film thickness difference, as shown in Supplementary Fig. 5a. This flexible joint improved the adaptivity of the feet for complex surfaces. After the assembly process of the robot, as shown in Supplementary Movie 1, the consequent prototype weighed approximately 19.7 g (the soldered one) and was about 970 μm in height at minimum, excluding wires.

As shown in Supplementary Fig. 5b, c, three independent high-voltage sinusoidal suppliers (HV_1, HV_2, and HV_3) with the same voltage amplitude (zero to peak) and variable frequencies were used to drive the 2-DOF actuator, with an initial phase difference of 120° between the three voltage suppliers, while another squared power supplier HV_4, powered the electrostatic adhesive feet. The system was controlled by a microcontroller (Arduino Uno) via solid-state relays (Panasonic AQV 258), as detailed in Supplementary Note 2. The electrical wiring of the robot is shown in Supplementary Fig. 5d. $A_{1\text{-}X}$, $A_{2\text{-}X}$, and $A_{3\text{-}X}$ are used to drive the actuator for motion in the X direction, while D_X controls whether the X-direction actuation module is in the released or adhered state. A similar wiring applies to the Y direction. As illustrated in the sequence diagram in Supplementary Fig. 5e, the voltage phase orders for the X and Y directions can be changed on purpose. There is an approximately 50 ms time interval during the switching between different modules or motion directions. By regulating the on/off status of the relays, we can control the 2-DOF moving direction of the actuator in the plane. By cooperating with the 2-DOF actuators and the adhesive feet, the robot can be controlled to travel on a complex surface or conduct the machining tasks with one actuator fixed on the ground and drive the other actuator to move and follow prescribed trajectories.

Experimental characterization of the 2-DOF electrostatic film actuator

We experimentally verified the displacement accuracy, speed, trajectory control, and output force. In the displacement measurement setup shown in Supplementary Fig. 6a, the stator module was fixed onto an acrylic plate, while two reflectors were attached at 0° and 45°, respectively, to the slider module for position measurement using a laser displacement sensor (HG-C1200, Panasonic). The $V_{0\text{-}p}$ (zero-to-peak voltage) of power suppliers (HV1, HV2, and HV3) was set to 450 V. The stroke of the actuator is 100 mm (58.8% of the length) in both X and Y directions.

We evaluated the actuator's cyclic durability and displacement across driving frequencies. The actuator operated for more than 10,000 cycles with a 20 mm stroke at a 20 Hz driving frequency, as shown in Supplementary Fig. 6b. The non-linearity error of the travel distance was as high as 0.23%, as shown in Supplementary Fig. 6c, with a 10 Hz driving frequency. Thus, the movement of the actuator can be accurately controlled by open-loop (estimating the distance just by operation duration). Moreover, the speed of the actuators was linearly proportional to the driving frequency (non-linearity error 3.46%), as estimated by the theory. The high linearities of the displacement and speed were also true for the composition of the X and Y movements. For example, the 45° displacement and speed display 0.13% and 2.29% non-linearity error, respectively, in our experiments (Fig. 2d, e). Moreover, the number of actuation interfaces in the actuators also did not affect the speed. The maximum speed of the actuator was approximately 290 mm/s with a driving frequency of 240 Hz.

We measured the linear positioning performance of the actuator by controlling the actuator (10 Hz driving voltage) to move reciprocally with a distance of 12 mm along 0°, as shown in Supplementary Fig. 6c, with the calculation method detailed in Methods and based in part on the International Organization for Standardization (ISO) 230-1 and 230-2[52,53]. The mean bidirectional positioning accuracy was calculated to be 23.1 μm (merely 0.023% of the actuator's full stroke of 100 mm), with bidirectional repeatability of 21.7 μm, as shown in Supplementary Fig. 6d and detailed in Methods. These indices are comparably high and acceptable for industrial machining applications, considering that the accuracy and repeatability of industrial robots are approximately 0.1–1.0 mm and 0.03–0.3 mm, respectively[54]. These indices can be further improved in the future by optimizing the structure design and control algorithms[55]. The step response to a 10 mm target position at a driving frequency of 100 Hz is shown in Supplementary Fig. 7, with a settling time of 83 ms and a root-mean-square error (RMSE) of 0.31%.

The actuators can follow prescribed planar trajectories, and this capability is necessary for machining. Here, we demonstrated that the actuators moved with trajectories of rectangles, circles, and complex trajectories (the outline of the letters of SUSTech, the authors' affiliation), with driving frequencies of 30 Hz, 10 Hz, and 2 Hz, respectively (Fig. 2f, g, Supplementary Movie 2). The RMSE increases to above 0.154 mm, particularly for trajectories with sharp turns (detailed in Supplementary Table 6). These errors resulted from simply shutting down or turning on the three-phase sinusoidal voltages by relays and can be improved in the future by advanced control methods and more sophisticated driving circuits. A sub-centimeter 2 DOF electrostatic actuator with actuation films of 9 × 5 mm^2 footprint was fabricated and demonstrated planar motion under 2 Hz driving, as shown in Supplementary Fig. 8 and Supplementary Movie 3.

The output force of the 2-DOF electrostatic film actuator was measured by the setup shown in Supplementary Fig. 9a. The stator module was fixed, while the slider module was horizontally connected to the force sensor (LSB201, FUTEK) via a spring. The slider module was placed on a sliding rail to reduce friction during the test. As the slider module moved leftward (powered by different voltage amplitudes with a 10 Hz driving frequency), it stretched the spring, and the sensor recorded the increasing force in real time. Based on Eq.(1), the model of force can be simplified to $K = k_F A_e = 1.4\times10^{-6}\,N{\cdot}V^{-2}$, and a constant term b is presented due to the friction force between every film. The maximum driving force was acquired when the slider module could no longer stretch the spring. The maximum output force of the actuator with five actuation interfaces was as high as 1.07 N at $V_{0\text{-}p}$ = 600 V (Fig. 2h). The data also show that the output force of the

actuator was proportional to the number of actuation interfaces, as estimated. The exception occurred when the $V_{0\text{-}p}$ exceeded 500 V, probably due to air breakdown and subsequent discharge[56]. Moreover, we measured the output force of the actuators in different directions (X direction, Y direction, and 45° direction, respectively) by changing the sensor directions (Supplementary Fig. 9b). As predicted by Eq. (1), the resultant force at 45° was the composition of the force components in the X and Y directions, as shown in Supplementary Fig. 9c. The discrepancy probably resulted from the effect of the unfavored torque in the measurement of 45° force.

Another feature of the actuator is that it can move while bending. In the experimental setup shown in Fig. 2i, the actuator was bent by a jig and then driven (at $V_{0\text{-}p}$ = 450 V and 10 Hz driving frequency) to pull the force sensor via a spring. As shown in Fig. 2i, the single-interface output force remained stable until the bending radius fell below 50 mm, whereas the five-interface configuration degraded faster, likely due to interlayer interactions in the stacked films. The stacked actuator still generated a certain output force under most bending conditions.

Characterization of the electrostatic adhesive pad

The adhesive friction force of the foot pads was measured by pulling the adhesive pad on the substrates by a force sensor (LSB203, MTS) on a tensile test machine (Supplementary Fig. 10a). Due to different adhesive mechanisms on conductive and insulating substrates (based on free charges and polarities, respectively)[57], we powered the adhesive pads with different voltages corresponding to different substrates. For conductive substrates (e.g., the aluminum plate), the interdigitated electrodes were all connected to the same power supply, while the two-phase electrodes for insulating substrates (e.g., the rubber sheet) were connected to different voltages. The models for tangential adhesive force were derived based on Eq. (5). As shown in Fig. 2j, on the conductive substrate, the driving frequency had little effect on the adhesive force (with a maximum value of 4.87 N at ±450 V and 50 Hz driving frequency), but DC (direct current) voltage significantly reduced the output force (less than 32.2% of the force generated by the AC voltage), probably since the charges accumulated on the surfaces blocked the coulomb force. Similarly, DC voltage also resulted in a significantly low adhesive force on the rubber substrate. Moreover, on the rubber sheet, as shown in Fig. 2k, the driving frequencies of the square wave caused different adhesive force amplitudes, in which a 50 Hz driving frequency generated a higher force. Although the adhesive force on the rubber sheet was much lower than that on the aluminum plate, the breakdown voltage was higher on the rubber sheet. Thus, the force could reach 8.65 N at ±1000 V, while the air would break down if the voltage exceeded ±500 V on an aluminum substrate.

Furthermore, electrostatic adhesion can be extensively applied to vast material types. At ±1400 V and a 50 Hz driving frequency, the maximum adhesive forces on paper, PLA, acrylic, and wood were approximately 1.87 N, 1.02 N, 0.787 N, and 0.157 N, respectively (Fig. 2l). At ±450 V and a 50 Hz driving frequency, the lateral adhesive force measured on textured brushed aluminum and dusty aluminum was 3.84 N and 0.339 N, respectively (Supplementary Fig. 10b). At ±1400 V and a 50 Hz driving frequency, the lateral adhesive force measured on fibrous cotton[58] and porous polyurethane (PU)[59] was 0.042 N and 0.031 N, respectively. The result shows that dusty, porous, and fibrous substrates caused substantial degradation of adhesion (with the largest reduction of 98.3% on the PU substrate compared with paper). Surface roughness tests were conducted using 45# carbon steel coupons (20 × 23 mm) with seven Ra values (0.1–6.3 μm). At ±450 V and 50 Hz, the adhesive force decreased monotonically with Ra (Fig. 2m), dropping from 0.733 N at Ra = 0.1 μm to 0.072 N at Ra = 6.3 μm (90.1% reduction). At Ra = 0.4 μm, comparable to typical aircraft surface roughness[60] (Ra between 0.4 and 0.5 μm), the adhesive force remained 0.557 N (24.0% reduction), indicating acceptable adhesion on aircraft-like surfaces.

The flexible adhesive pad allows adhesion on complex surfaces. Here we measured the adhesive force on curved surfaces of different radii, as shown in Fig. 2n. At ±450 V and a 50 Hz driving frequency, the adhesive force shows no noticeable change for surface radii larger than 400 mm. The adhesive force on the concave surface decreased by 75.4% (from 5.35 N to 1.32 N) as the surface radius was reduced from 1000 mm to 50 mm (comparable to the 50 mm pad width), and the adhesive force was still strong enough to support the robot body (19.7 g). Moreover, the adhesive force on the convex and concave surfaces of the same radius remained close. Thus, the adhesive pad allows the fixture of the robot on complex surfaces.

Experimental characterization of the robot

Here, we demonstrated the robot's capabilities of locomotion (on planes and curved surfaces), payload, and robustness. By cooperating with the 2-DOF planar electrostatic film actuator and the electrostatic adhesive pads, the robot can travel between different positions on the ground in an inchworm-like locomotion manner, as shown in Supplementary Fig. 11. The voltage amplitudes for the actuator were set to $V_{0\text{-}p}$ = 450 V, and the adhesive pads were powered with a ± 450 V square voltage and 50 Hz driving frequency. We control the driving frequency of the supplier voltages of the actuator to adjust the speed of the robot. As shown in Figs. 3a, b (Supplementary Movie 4), the robot demonstrated a straight movement at a speed of 27.6 mm/s (0.102 body length per second, BL/s) and then turned an angle of 45° (39.0 mm/s, 0.144 BL/s) on an aluminum plate, at 80 Hz driving frequency. The robot demonstrated trajectory tracking (Supplementary Fig. 12) with a maximum absolute error (MAE) of 0.26 mm and an RMSE of 0.124 mm at a 50 Hz driving frequency.

When the actuator and adhesive pads were driven at 10 Hz, the voltage and current signals of the high-voltage amplifiers are shown in Fig. 3c. Power consumption of the actuator, adhesive pads, and the robot at different driving frequencies is summarized in Supplementary Fig. 13, showing that increasing the driving frequency of both the actuation module and the adhesive feet increases power consumption. The cost of transport (CoT) was calculated as described in the Methods and is shown in Fig. 3d. The CoT ranged from 15.3 to 41.7, reaching a minimum of 15.3 when the actuator and adhesive pads driving frequencies were 30 Hz and 10 Hz, respectively.

With a similar control strategy, the robot could move on various materials, such as paper, wood, rubber, and acrylic, at 20 Hz driving frequency, as shown in Fig. 3e, although the locomotion speeds were slower since the lower adhesive force on the insulating substrates resulted in more slippage.

The compliant robot can also move on curved and vertical surfaces, which are common in large structures such as airplanes and ships. As shown in Fig. 3f, the robot climbed on the curved surface (560 mm radius) at 50 Hz driving frequency, while both the actuator and adhesive pad passively adapted to the complex surface and kept an intimate contact. Without any changes to the control strategy, the robot could also move on a vertical surface, as shown in Fig. 3g. With its flexible structure, the robot can be compliant with external rigid frames to adapt to substrates of different geometries. As shown in Fig. 3h, the robot achieved locomotion on the outer surface of a cylinder with a circumference equal to its body length (270 mm), corresponding to an outer radius of 43.0 mm, under 10 Hz driving. This corresponds to a circumferential dimension-to-radius ratio of 6.28, which is the largest among the robots listed in Supplementary Table 3, highlighting strong geometric conformability. Similarly, with an alternative rigid frame (mass of 22.5 g), the robot locomoted on an inverted curved surface with a radius of curvature of about 600 mm (Fig. 3i), although the rigid frame may limit its capability to navigate inverted surfaces with higher curvatures.

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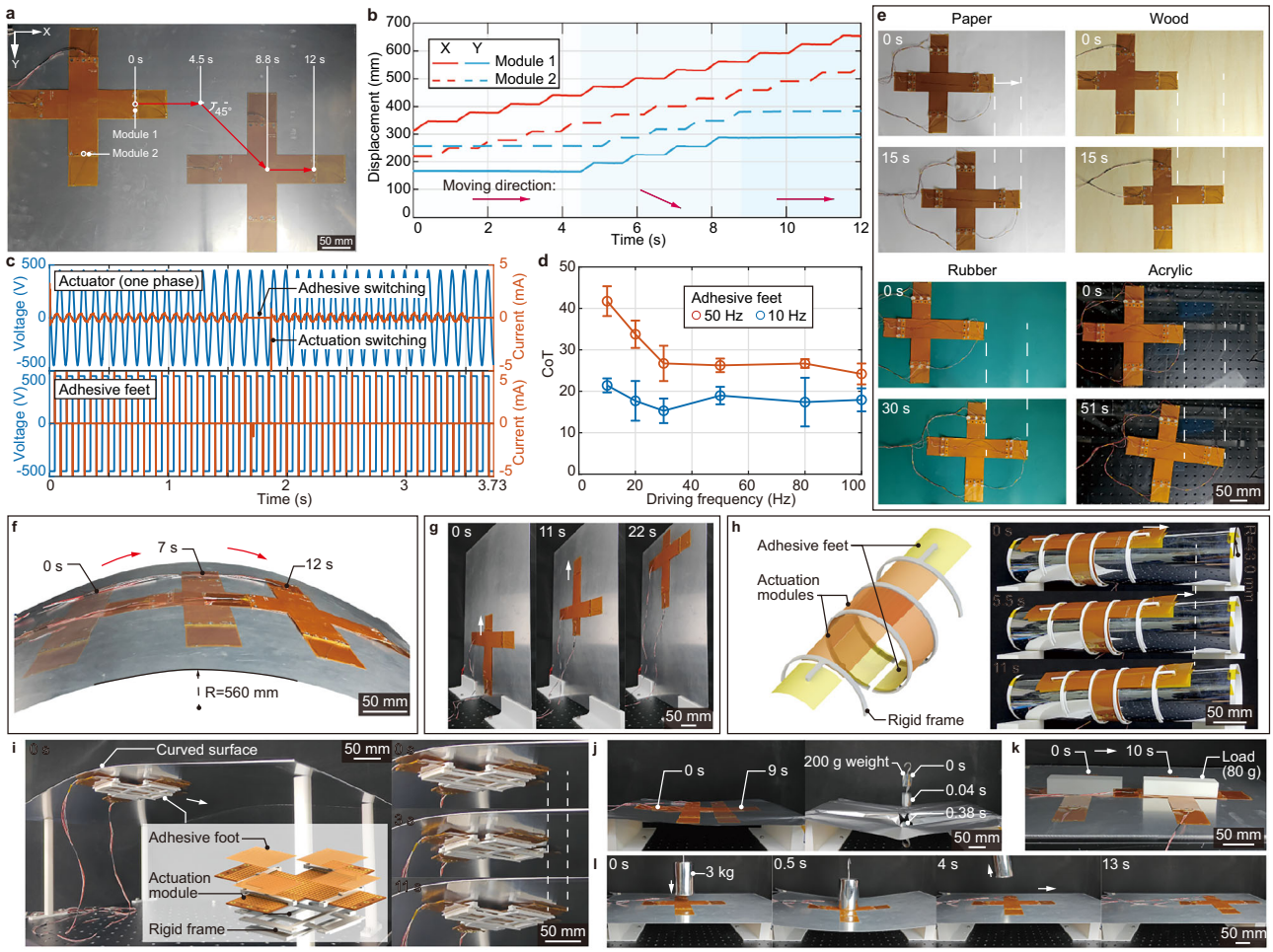


Fig. 3 | Performance demonstrations of the robot. a Planar movement on a conductive substrate (an aluminum board). **b** The displacement during the planar movement. **c** Output voltage and current signals of the high-voltage amplifiers. **d** The cost of transport (CoT) values of the robot under different driving frequencies. **e** Locomotion on various insulating substrates, including paper, wood, rubber, and acrylic. **f** Traversal over a curved surface. **g** Vertical surface climbing. **h** Locomotion on a cylindrical surface (circumference: 270 mm, equal to the robot's length) with a rigid frame. **i** Locomotion on an inverted curved surface with a rigid frame. **j** The robot safely traverses a fragile surface composed of 10 μm-thick aluminum foil, which can be easily torn by a 200 g weight. **k** Payload performance demonstration by bearing an 80 g weight, 4.06 times the robot's dead weight. **l** Robustness test under an external impact generated by a 3 kg weight (152 times the robot's weight). The data are presented as mean values ± SD for the number of trials $n = 5$.

Characterized by its lightweight, the robot was able to move on fragile surfaces. As shown in Fig. 3j, the robot successfully traversed a fragile surface made of 10 μm-thick aluminum foil, suspended between two adapters. This surface was so delicate that even a 200 g weight was sufficient to tear it apart. Although the robot had a thin profile, the actuation force of the robot was able to carry a payload (80 g) of 4.06 times the dead weight to move on an aluminum plate at 50 Hz driving frequency, as shown in Fig. 3k and Supplementary Movie 5. Due to the compliant actuators and adhesive pads, the robot is robust to external impact. As shown in Fig. 3l (Supplementary Movie 6), a 3 kg weight (152 times the robot's weight) was dropped onto the robot. Both the supporting aluminum plate (1 mm thick) and the robot were impacted and consequently bent largely. After the weight was removed, the robot recovered due to its reliability and survived the impact at 50 Hz driving frequency. Moreover, this demonstration also shows that even on a thin plane, which is common on planes and ships that are built with thin shells, the lightweight robot does not result in any strain and stress on the structure, which a previous heavy in-situ robot (usually heavier than 7 kg) might cause surface or structural damage.

Demonstrative applications of in-situ inspection and machining

Here, we demonstrate the robot in three representative tasks, including inspection, grinding, and EDM. The increasing demand for in-situ machining and inspection of large workpieces has attracted the development of machine robots as an alternative to traditional bulky and rigid machine tools. While wheeled and legged platforms have demonstrated mobility and adaptability, current systems remain heavy (over 7 kg), bulky (minimum dimension over 0.35 m), as listed in Supplementary Table 4, and are primarily limited to flat and external spaces. However, complex and confined (narrow sub-centimeter gaps) spaces are common in aircraft and ships[61,62].

For narrow gap inspection (Supplementary Movie 7 and Fig. 4a), a micro camera (OV9734, OmniVision Technologies Inc.) was mounted on the robot's foot. The robot was assigned to pass through a 7 mm gap with logos printed on the inner top surface (Fig. 4b). During the inspection, the 2-DOF actuator was driven at 10 Hz, resulting in a slow and stable speed (approximately 12 mm/s) to capture clear images. Moreover, inside the gap, due to the 2-DOF actuation capability, the robot did more than move forward linearly. Upon reaching the marked logo position, it executed a deliberate lateral (leftward) movement driven solely by the slider, performing a horizontal scan to fully capture the visual content. Once the scan was complete, the robot resumed forward locomotion. To further demonstrate its applicability in inspection, the robot was deployed in a gap with ribbed slabs (which is common in airships[63]) at the top (Fig. 4c), where a matrix scanning trajectory was executed while capturing video through the onboard

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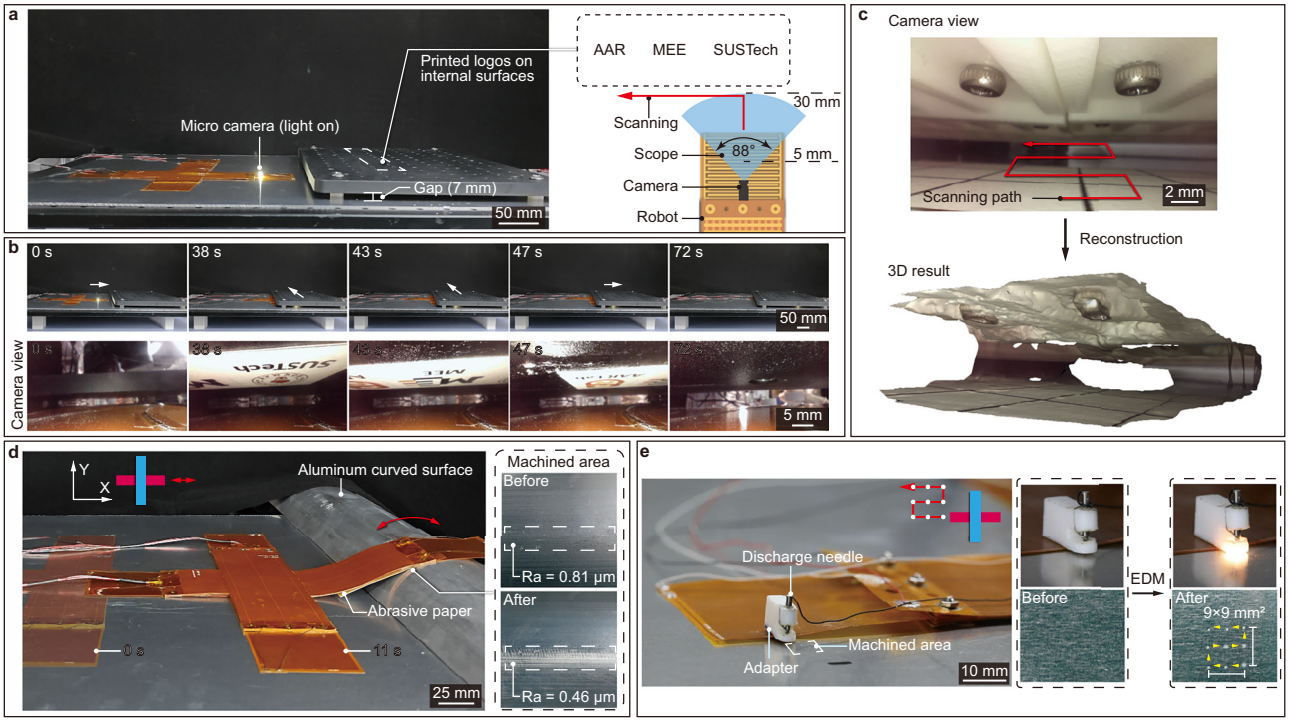


Fig. 4 | Demonstrative applications of the robot. a The narrow gap inspection scenario and the robot setup. **b** Motion images of the robot and inspection images captured by the micro camera. **c** Scanning and reconstruction of internal structures within a gap with ribbed slabs. **d** Grinding demonstration on a curved surface, with surface images and arithmetic average roughness (Ra) values before and after processing. **e** Electrical discharge machining (EDM) operation, showing the robot in action and the processed area before and after machining.

camera. Consequently, the 3D structure of the inner environment of the gap was reconstructed (detailed in Methods) by extracting the representative frames for further evaluation and analysis. The robot's ability to perform in-gap scanning operations in addition to linear locomotion highlights its precise planar maneuverability and suitability for inspection in confined environments.

In-situ machining in narrow spaces on large structures is also required[62]. For example, traditional robots for flexible grinding are large and rigid, unavailable for operation in confined spaces or traveling on complex surfaces[64,65]. However, the small height of the robot facilitates its trespass through the small openings, and the flexibility enables its adaptation to various irregular surfaces. Attached with a 1000-grit abrasive paper to the bottom (as shown in Fig. 4d), the robot was able to grind (Supplementary Movie 8). After the robot reached the target position at 50 Hz driving frequency, the Y-direction actuation module and electrostatic adhesive feet adhered to the grounded aluminum plate, while the X-direction module with abrasive paper executed reciprocating grinding on the target surface for 10 min at 30 Hz driving frequency, as shown in Supplementary Fig. 14a. The surface roughness was measured by a roughness tester (TR200, Beijing Time Technologies Co. Ltd), as shown in Supplementary Fig. 14b. The Ra value of the grinding area was 0.81 μm before grinding and 0.46 μm (reduced by 43.2%) after grinding, as shown in Fig. 4d and Supplementary Fig. 14c, confirming that the robot achieved flexible grinding on curved surfaces. To enable selective grinding, an electrostatic adhesive pad with abrasive paper was mounted on the side of the adhesive foot. With the pad voltage off, the abrasive paper remained non-contact with the substrate during locomotion to avoid unintended damage and was switched to voltage on at the target region to engage grinding, producing a clear abraded area after 20 cycles (Supplementary Fig. 15 and Supplementary Movie 8).

EDM is an extensively adopted machining method for metal material removal, but traditional EDM machines are large and non-portable, making it challenging for EDM in a confined space[66]. However, with our ultrathin robot, EDM can be conducted in a gap or on a curved surface. As shown in Fig. 4e, we attached a tungsten discharge needle (0.05 mm in diameter) to the robot foot with a 3D-printed adapter. After traveling to the aim position, one module of the 2-DOF actuator was fixed at the ground by the adhesive pad. Then the other module could be driven to move freely. Here, the free actuation module carried the needle to move and stop at the nine points in the 3×3 array one by one (Supplementary Movie 9). At each point, the aluminum material was successfully removed by EDM (± 1000 V, 1 Hz).

Discussion

This study presents a compact and flexible robot integrating the 2-DOF electrostatic film actuator with electrostatic adhesive pads. The resulting system features a thin profile (970 μm) and low weight (19.7 g), while demonstrating robust mobility across a wide range of substrates, including both conductive and insulating surfaces, on flat, vertical, and curved geometries. The robot achieves high motion performance, including a positioning accuracy of 23.1 μm, repeatability of 21.7 μm, maximum speed of 290 mm/s, and a superposable output force of 1.07 N, all within a compact and compliant form factor. The robot supports payloads exceeding 4.06 times its own weight and has endured a 3 kg impact without loss of functionality, evidencing mechanical resilience. Its capability was further validated in practical in-situ scenarios, including narrow-gap (7 mm) inspection and 3D reconstruction, flexible grinding on curved surfaces, and localized EDM. These demonstrations showcase the robot's potential to operate in geometrically constrained and sensitive environments where traditional rigid robots are ineffective.

Future iterations of the system will incorporate an onboard high-voltage amplifier, microcontroller, communication unit, and power supply to eliminate tethered constraints. High-permittivity dielectric layers will be adopted to enhance adhesive-pad performance and expand applicability across a wider range of substrates. Reducing the thickness of the actuation film will be explored to improve

conformability in multiple film stacking. In addition, integrating a compact variable-stiffness design[67] is expected to improve conformability on tight curvatures and avoid buckling during inverted locomotion without an external rigid frame. Moreover, integrating the miniature sensor[44] or the self-sensing method[55] will provide position feedback for closed-loop control, enabling a PID positioning servo controller[68] and supporting future exploration of sliding mode[69] or neural network-based[70] controllers to further improve positioning accuracy.

Methods

Fabrication of the actuation films for the electrostatic film actuator

The double-sided copper-clad laminate (composed of a 25 μm polyimide base layer and two 18 μm copper layers, AP8515R, Pyralux) was used as the substrate of the 2-DOF electrostatic film actuator. Negative-type dry films were laminated on both sides of the substrate. Ultraviolet light was then projected through an exposure mask containing the designed circuit patterns. The exposed area of the photosensitive dry films hardened, forming resist layers with low solubility, while the unexposed areas were dissolved using a developing solution (1% Na_2CO_3). The uncovered copper areas were etched by etching solution ($CuCl_2$ and HCl), leaving only the circuit areas intact. After etching, the hardened dry films were stripped with caustic soda (3% NaOH), revealing the patterns of the bare flexible circuit. For insulation, pre-formed coverlay (13 μm polyimide layer and 13 μm thermal adhesive layer, FR7001, Pyralux) was thermally bonded to the double-sided copper clad laminate using a heat press at 185 °C, 2.3 MPa for 3 h.

To make the 2-DOF actuators, the actuation films were bonded on the two edges by melted solder for a small thickness of the actuators. This bonding method resulted in poor maintenance. Thus, in this work, we also connected the films with M2 screws and nuts for assembly and disassembly convenience.

Performance characterization of the electrostatic film actuator and the electrostatic adhesive pad

To demonstrate the performance of the 2-DOF electrostatic film actuator, experiments were conducted to evaluate its output force, positioning accuracy, and motion speed. In these experiments, control signals were generated by a control board (SCB-68A, National Instruments) and then amplified 1000 times by high-voltage amplifiers (Model 615-3, Trek) to drive the electrostatic film actuator. The output force was measured using a force sensor (LSB201, FUTEK), while the position of the slider was recorded using a laser displacement sensor (HG-C1200, Panasonic). For the experiments testing the performance of the electrostatic adhesive pad, the experimental setup is shown in Supplementary Fig. 10, a force sensor (LSB203, MTS) was fixed on a tensile test platform (Model 42, MTS) and moved upward with a velocity of 50 mm/min.

Sensor data was collected by analog signal acquisition boards (TB-4309 and TB-4330, National Instruments) in the experiments of the electrostatic film actuator. The sampling time was set to be 1 ms for each measurement. The mean and SD force results were calculated from 7 measurements.

Positioning accuracy and repeatability

The measuring range for one direction of each continuous reciprocal motion cycle was divided into ten segments in this work. To obtain bidirectional results, the data at the endpoints of the measuring range were excluded. Consequently, the positioning errors in the positive direction, denoted as $x+_i$ (i = 1, ..., 8), and in the negative direction, denoted as x_{-i} (i = 1, ..., 8), were calculated as the differences between the actual and target positions at eight intermediate points. For multiple reciprocal cycles (n cycles), the positioning errors could be represented as x^+_{ij} and x_{-ij} (j = 1, ..., n). For each position, the mean unidirectional positioning errors were then calculated as follows:

$$\bar{x}_i^+ = \frac{1}{n}\sum_{j=1}^{n} x_{ij}^+, \tag{6}$$

$$\bar{x}_i^- = \frac{1}{n}\sum_{j=1}^{n} x_{ij}^-, \tag{7}$$

where n = 6 for this work. The mean bidirectional positioning error at each position was calculated:

$$\bar{x}_i = \frac{\bar{x}_i^+ + \bar{x}_i^-}{2}. \tag{8}$$

The standard deviations at each position were calculated:

$$s_i^+ = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(x_{ij}^+ - \bar{x}_i^+)^2}, \tag{9}$$

$$s_i^- = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(x_{ij}^- - \bar{x}_i^-)^2}. \tag{10}$$

The bidirectional positioning accuracy was calculated:

$$A = \max[\bar{x}_i^+ + s_i^+; \bar{x}_i^- + s_i^-] - \min[\bar{x}_i^+ - s_i^+; \bar{x}_i^- - s_i^-]. \tag{11}$$

The bidirectional positioning repeatability R_i for each position was calculated:

$$R_i = \max[s_i^+ + s_i^- + |\bar{x}_i^+ - \bar{x}_i^-|; 2s_i^+; 2s_i^-], \tag{12}$$

and the maximum bidirectional repeatability $R = \max[R_i]$.

The trajectory errors in Supplementary Table 6 were calculated based on the Euclidean distance between the target trajectory points and the actual trajectory points.

Cost of transport

The CoT is calculated as:

$$\mathrm{CoT} = \frac{P_{\mathrm{avg}}}{m_{\mathrm{r}} g v_{\mathrm{avg}}}, \tag{13}$$

where P_{avg} is the average power consumption of the robot (two actuation modules and four adhesive feet), m_{r} is the robot mass, g is the acceleration due to gravity, and v_{avg} is the average locomotion speed.

3D reconstruction

To ensure robust reconstruction, keyframes were extracted from the robot's recorded video with at least 70% overlap between consecutive frames. These keyframes were then imported into Meshroom for processing. In the CameraInit node, the camera type was set to radial3, with a focal length of 30 mm, a pixel ratio of 1, and an image resolution of 1280×720 pixels. Following initialization, the standard Meshroom pipeline was executed, which sequentially applied FeatureExtraction, ImageMatching, FeatureMatching, StructureFromMotion, PrepareDenseScene, DepthMap, Meshing, MeshFiltering, and Texturing, resulting in a complete 3D reconstruction of the scene.

Data availability

All data generated in this study are included in the article and its Supplementary Information. Source data are provided in this paper. Source data are provided with this paper.

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Author contributions

H.W. and H.W. jointly conceived the machine robot. H.W. designed and conducted all experiments, collected and analyzed the experimental results, and wrote the manuscript. Y.D., T.W., and L.S. analyzed the finite element model of the electrostatic film actuator and assisted in the fabrication of the electrostatic film actuator. W.W., Z.Z., and J.Z. assisted in the assembly of the robot. H.W. reviewed and edited the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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